







different cameras. This would have, however, not happened if the field of view changed in more drastic ways. This scenario would likely require adaptations to the image pre-processing steps, the neural network architecture and the training dataset. In this study, the field of view (45 degrees) was within the fields of view presented during training, and thus DR lesions have similar relative sizes across different images. Additionally, the smartphone-based fundus camera has been validated against standard desktop systems for image quality.<sup>4,14</sup> This has specifically shown that DR grading by experts is comparable on the systems.

Moderate NPDR is the cut off for detecting RDR as per the International Council of Ophthalmology guidelines for screening of DR and the AAO preferred practice patterns.<sup>7,15</sup> Accordingly, this was also the threshold used for the AI to trigger referral. When the AI results were compared to the clinical grades for RDR, the sensitivity was 100% and specificity was 65.2%, respectively. The specificity was lower than that reported in previous validation studies (86.73–92.5%) on the smartphone-based system.<sup>7,16,17</sup> On further analyzing the low specificity, we found that there were eighteen cases of mild NPDR and five cases of no DR on clinical assessment that were detected by the AI algorithm as RDR. Interestingly, when the consensus image grading of the same mild NPDR patients were cross verified, fifteen were graded as RDR, with two of them graded as STDR.

On analyzing the five no DR cases on clinical exam that were picked as RDR positive by the AI, two had a consensus of any DR, with one of them being RDR too on image grading. We re-examined the class activation maps on these five subjects and found that three subjects had other lesions – drusens (in two) and Pigment Epithelial Detachment (in one) that triggered the AI to give a positive result.

The analysis of the spuriously low specificity of the RDR algorithm against clinical exam also showed that the kappa for clinical exam (Cohen's kappa 0.72) was lower, compared to the agreement obtained by experts during image grading (Cohen's kappa 0.89 and 0.86). The variation found was higher in milder stages of disease. Literature indicates a wide range of interobserver and grader reliability, ranging from 0.22 to 0.91.<sup>18</sup> We found that the consensus image diagnosis from two experts was more consistent and reliable than a single observer clinical evaluation. This further justifies Krause et al's interpretation where they found that majority decision to have a higher sensitivity than any single grader.<sup>19</sup> Most of the images (15/18 eyes) that were graded as mild NPDR on clinical exam were graded as moderate NPDR or more severe disease on image grading. Thus, specificity went up considerably to 83.7% on image grading with multiple graders. Well known clinical trials like the ACCORD and FIND have also found image grading to be superior to clinical grading to detect early to moderate changes in DR over time.<sup>20</sup> This is also backed by regulatory authorities like FDA who advocate for image grading by multiple certified graders on a consensus or adjudication basis. While clinical assessment provides an opportunity to examine the entire retina, three field imaging with multiple graders provided sufficient information for reliable screening to detect RDR.

While we found the results to be comparable to our previous studies using the smartphone-based camera, the modest increase in sensitivity and decrease in specificity<sup>4,16,17</sup> is possibly due to minor variations expected in image sharpness. The decrease in specificity is primarily due to an overcall of mild NPDR cases. A desktop camera like Topcon has better sharpness with mild lesions being more prominent and hence more likely to be picked up by the AI.

The Medios AI system consists of two components: an AI for image quality analysis and an AI for referable DR. This allows the operators to get live automated feedback at the time of image capture. It enables the user to understand whether the image captured is of sufficient quality or needs a recapture. This image quality algorithm has been optimized for use on the Remidio FOP-NM10 device. In this study, we assessed the performance of the AI quality check on the images captured with Topcon camera using the same 0.2 threshold that was used on the original version of the system (deployed on Remidio FOP). The sensitivity of detecting ungradable images was 100% and the sensitivity to detect gradable images was 82.8%. We found that an improved performance can be achieved on the Topcon system by setting the threshold at 0.5. The sensitivity for detecting gradable images improved to 89.0% with a sensitivity drop to 96.15% for detecting ungradable images. Given the minor differences in the sharpness of their imaging system, the threshold of the algorithm will require to be varied prior to deployment on a new camera system.

The strengths of this study are post-hoc analysis on a dataset with good representation of all stages of disease, simultaneous comparison of the AI performance to two reference standards (image grading and clinical assessment) as well as an assessment of the image quality algorithm.

This study has some limitations. First, the AI has been tested with images with similar fields of view. The performance of the AI models when deployed on images with a significantly different field of view needs to be assessed. Second, the images were analyzed using the same AI model as deployed offline on the Remidio FOP, but on a Cloud Virtual Machine. The performance of the system after a future offline integration of the models on a Topcon Fundus camera system will require further study.

### Conclusion

To the best of our knowledge, this study is the first of its kind to compare an AI-based screening algorithm for DR to both clinical examination and consensus image-based grading. This study adds to the growing evidence on image-based grading being more consistent and reliable for screening DR than a clinical exam. The AI which had previously been validated only on a smartphone-based fundus camera showed a high sensitivity and specificity in screening for RDR and any stage of DR on images captured on a standard desktop camera. This indicates that this algorithm can be used on both a high-end desktop fundus camera like Topcon and a smartphone-based system to screen for DR given the diversity in training dataset. Thus, it is a positive move towards a device-agnostic application of the AI for expanding the use in screening for DR in different settings. Further studies need to be conducted to assess the efficiency of the system on images from other cameras. This may provide a big boost in reducing the huge economic burden posed by DR globally.

### Disclosure

Divya Parthasarathy Rao, Anand Sivaraman and Florian M Savoy are Employees of Remidio Innovative Solutions. Medios Technologies, Singapore, where the AI has been developed, and Remidio Innovative Solutions Inc. USA, are wholly owned subsidiaries of Remidio Innovative Solutions Pvt Ltd, India. Dr Sabyasachi Sengupta reports personal fees from Novartis, India, Bayer, Intas, Allergan, outside the submitted work. The authors report no other conflicts of interest in this work.

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